

Some studies in facial expression recognition using deep learning

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Problem from a preschool

How do the kids feel in class?



- Happy?
- Angry?
- Sad?
- ...



- 1) How to recognize kid's face expression?
- 2) How to help the preschool to evaluate the education?

Tasks to do

- Data acquisition
- Data preprocessing
- Learning model to read kid's face feelings

Data collection



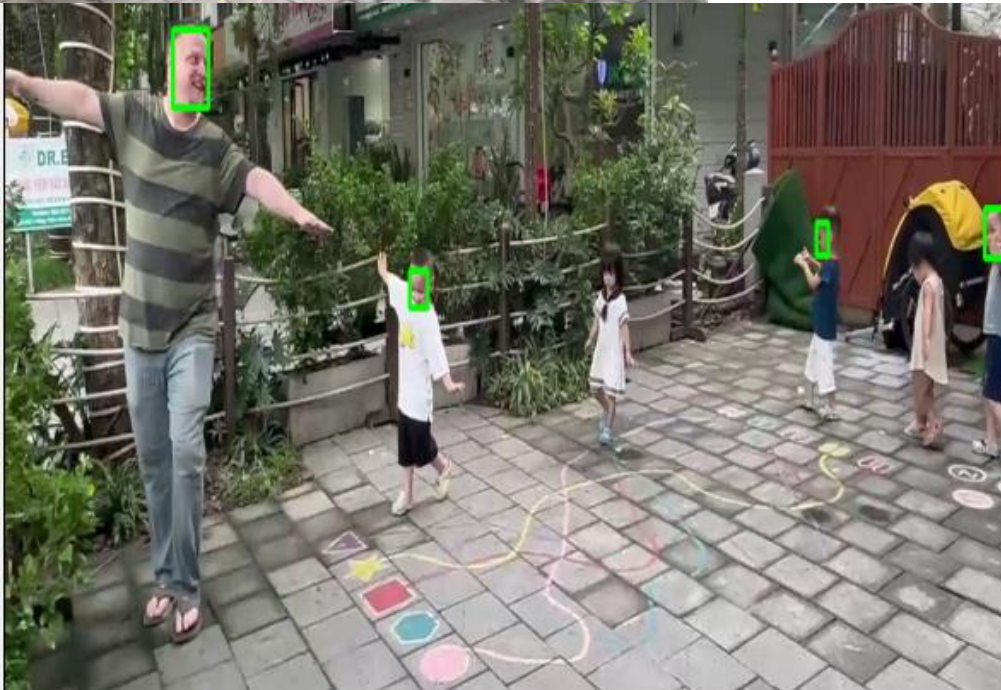
DATASETS

- Permission to access datasets
- >60 kids and teachers engaged
- Activities: Exercise, Math, Arts, Language, Music, outdoor

Face extraction

Data collected:

- Devices: Cameras(fixed or portable), smartphones
- Distances



Problem of data preprocessing

- How to extract as many faces as possible?
- How to rescale images from different distances?
- How to label face's feelings from the big datasets of preschool videos for learning?

Face classification



Activities: Exercise, Math, Arts,
Language, Music, outdoor,

Happy



Angry



Sad



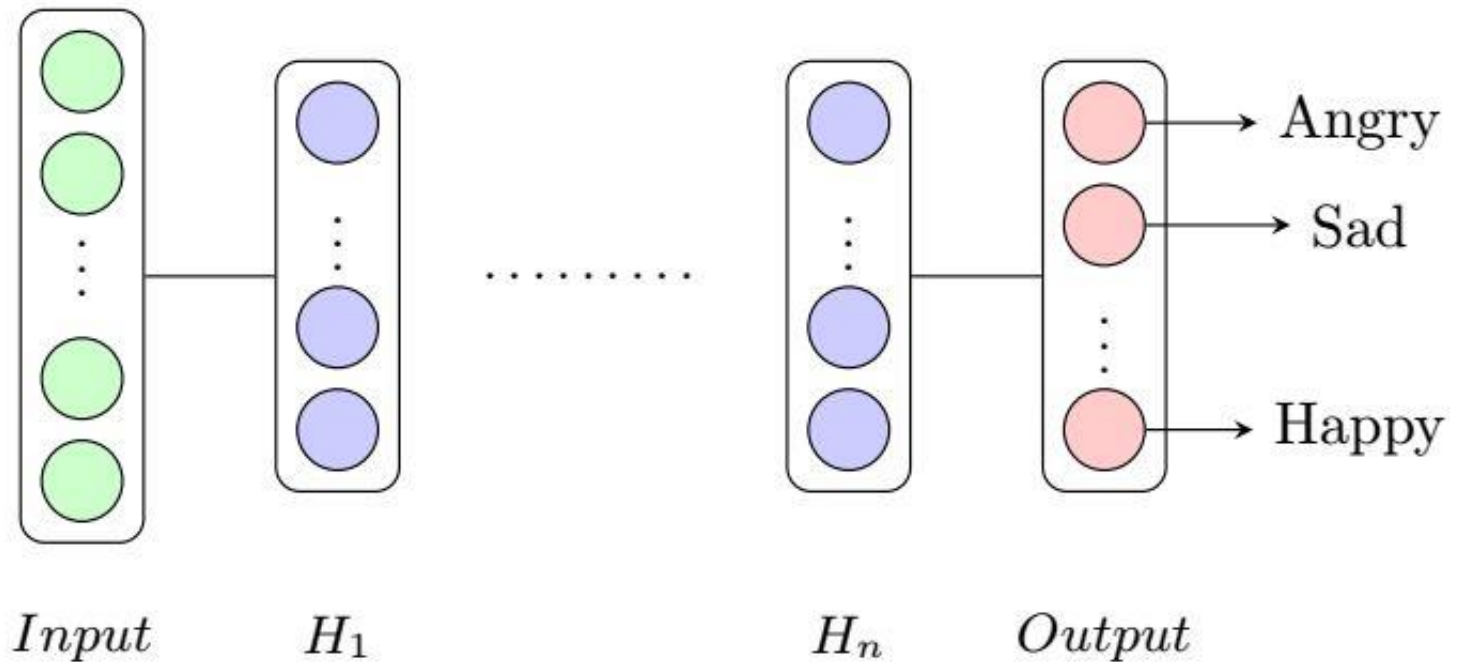
Learning Models

Datasets



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Deep learning model



Deep Learning Models of Interest

- Convolutional Neural Network
- Deep Boltzmann Machines
- AffectNets
- MTCNN
- FaceNets
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Thank you very much